MIL-C-50228(MU)
11 March 1969

MILITARY SPECIFICATION

CASE, TRANSIT, FOR TEST SET, T4090

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of Case, Transit used to protect T4090 test set in shipping and storage.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-B-2427 - Box, Ammunition Packing, Wood, Nailed
MIL-P-116 - Preservation, Methods of
MIL-T-4734 - Transit Cases, Combination Cases and Spare Parts Case for Ground Electronic Equipment USAF, Requirements for.
MIL-W-6858 - Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-resisting Alloys, and Titanium Alloys; Spot and Seam.
MIL-W-8604 - Welding of Aluminum Alloys, Process for.

F S C 8140

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8810353 - Riveting Instructions Aluminum and Aluminum Alloys Structural Assemblies
8824705 - Handle, Carrying
8834979 - Test Set, Atomic Weapon: T4090 w/Case
8834981 - Case, Transit
8834982 - Case Assembly, Bottom
8834983 - Case, Assembly, Top
8834986 - Case, Top
8834987 - Closure, Top
8834988 - Case, Bottom
ET-8834981 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Transit Case shall comply with all requirements specified on Drawings (dwgs) 8834979 without test set and marking and 8834981, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Vibration. The transit case (see dwg. 8834979) packed with a dummy test set (see 6.4) shall withstand vibrations without evidence of damage or loosening of parts when tested as specified in 4.4.1 and shall also comply with the pressure test specified on dwg. 8834981 after completion of the vibration test.

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3.5 Drop Test. The transit case (see dwg 8834979) packed with a dummy test set (see 6.4) shall withstand the free fall drop test MIL-P-116 without loosening of parts when tested as specified in 4.4.2 and shall also comply with the pressure test specified on dwg. 8834981 after completion of the drop test.

3.6 Low Temperature. The transit case (see dwg. 8834979) packed with a dummy test set (see 6.4) shall withstand exposure of minus 65 ± 5 degrees fahrenheit (°F) without evidence of cracking or deterioration of pads when tested as specified in 4.4.3. After completion of the test, the transit case shall comply with the pressure test specified in dwg. 8834981.

3.7 Paint Adhesion. (see dwg. 8834981) Paint adhesion shall be as specified in paint adhesion TT-C-490 when tested as specified in 4.4.4.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg. 8834979 without test set and marking) subassemblies (see Table I), and 1 set of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

Table I

Subassemblies submitted for first article inspection

Quantity	Drawing
1	8834982
1	8834983

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4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests need not be conducted in order specified in Table II.

Table II

First article inspections and tests

Examination or test	Contractor's inspection before submission	Destructive	Non-Destructive	Parts	Sub-Assemb	Assemb
1. Classification of characteristics (see 4.3.2.1)	X		X	X	X	X
2. Vibration (see 3.4)	X		X			X
3. Drop test (see 3.5)		X				X
4. Low temperature (see 3.6)	X		X			X
5. Paint adhesion (see 3.7)	X		X			X
6. Pressure (see 4.3.3.1)	X		X			X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table III

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96- 110	37
111- 130	38
131- 156	39
157- 190	40
191- 240	41
241- 340	42
341- 520	43
521-1120	44
1121 and over	45

Minor characteristic table IV

Lot size	Sample size
1- 25	100% inspection
26- 35	16
36- 45	17
46- 55	18
56- 65	19
66- 100	20
101- 200	21
201 and over	22

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Case, bottom (see dwg. 8834988 covering a detail of dwg. 8834982).

Categories	Characteristics	Method of inspection
Critical: None defined.		

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Major:
101. Length.....*Measure
102. Width.....*Measure
103. Height.....*Measure
104. Thickness, min.....Gage
105. Materials (see 3.1).....Visual

*Measure with suitable scale.

Minor:
201. Workmanship (see 3.8).....Visual/Manual

4.3.2.1.2 Closure, top (see dwg. 8834987 covering a detail of dwg. 8834983).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) available for part in accordance with dwg. 8834987.....Visual

Minor:
201. Workmanship (see 3.8).....Visual/Manual

4.3.2.1.3 Case, top (see dwg. 8834986 covering a detail of dwg. 8834983).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Length.....*Measure
102. Width.....*Measure
103. Height.....*Measure
104. Thickness, min.....Gage
105. Material (see 3.1).....Visual

*Measure with suitable scale.

Minor:
201. Workmanship (see 3.8).....Visual/Manual

4.3.2.1.4 Case assembly, top (see dwg. 8834983 covering a detail of dwg. 8834981) before applying epoxy and welding.

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Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Pitch diameter of thread	Gage
102.	Minor diameter of thread, max	Gage
103.	Location of threaded hole	Visual
104.	Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8834983	Visual
Minor:		
201.	Workmanship (see 3.8)	Visual/Manual

4.3.2.1.5 Case assembly, top (see dwg. 8834983 covering a detail of dwg. 8834981).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Welding	Visual
102.	Objective evidence (see 6.2) available for epoxy in accordance with dwg. 8834983	Visual
103.	All parts present and properly assembled	Visual
104.	Gasket groove smooth at joints	Manual
Minor:		
201.	Workmanship (see 3.8)	Visual/Manual

4.3.2.1.6 Case, transit (see dwg. 8834981 covering a detail of dwg. 8834979) after welding and before attaching stiffeners and catches, and painting.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Welding in accordance with MIL-W-6858 and MIL-W-8604 as applicable	Visual/Manual
102.	Top in line with bottom	Visual/Manual
103.	Location of hinge and relief valve	Visual
104.	No part missing, improperly located or assembled	Visual
105.	Threads staked on relief valve assembly	Visual
106.	Ends of hinges peened	Visual
107.	Sealing compound present	Visual

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108. Proper hinge operation.....Manual
109. Objective evidence (see 6.2) available for protective finish in accordance with dwg. 8834981.....Visual

Minor:
201. Workmanship (see 3.8).....Visual/Manual

4.3.2.1.7 Case transit (see dwg. 8834981 covering a detail of dwg. 8834979) after welding and before attaching stiffeners and catches.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. No paint on any rubber component.....Visual
102. Paint adhesion (see 3.7).....Visual/Manual

Minor:
201. Workmanship (see 3.8).....Visual/Manual

4.3.2.1.8 Case, transit (see dwg. 8834981 covering a detail of dwg. 8834979).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Rivet assembly in accordance with dwg. 8810353.....Visual
102. Location or assembly of any part.....Visual
103. No paint on catch and strike.....Visual
104. Sealing compound present on rivets.....Visual
105. No paint present on any rubber part.....Visual
106. Latch operation.....Manual
107. Handle operation.....Manual

Minor:
201. Marking.....Visual
202. No paint missing, including bare spot.....Visual
203. Workmanship (see 3.8).....Visual/Manual

4.3.2.1.9 Test set, Atomic Weapon: T4090 w/case (see dwg. 8834979) before closing cover.

Categories	Characteristics	Method of inspection

Critical: None defined.

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Major:
101. Pads tight in case.....Visual/Manual
102. Projection of top pad correctly located.....Visual
103. Dummy test set (see 6.4) not loose in pad.....Visual/Manual
104. No tear, split or crack or holes in pads.....Visual/Manual

Minor:
201. Workmanship (see 3.8).....Visual

4.3.2.1.10 Test set, atomic weapon: T4090 w/case (see dwg. 8834979).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201. Workmanship (see 3.8)		Visual/Manual

4.3.2.1.11 Preparation for delivery (see 5.2).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101. Packaging, packing and marking in accordance with MIL-STD-1169(MU)		Visual
Minor:		
201. Workmanship (see 3.8)		Visual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test classification
a. Pressure	Major
b. Handle	Major

These tests are not considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Pressure (see dwg. 8834981). Each transit case shall be subjected to this test (see 4.4.5). If the transit case fails to comply with the drawing requirement, it shall be classified defective and removed from the lot.

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4.3.3.2 Handles (see dwg. 8824705). Each handle shall comply with the applicable requirements of specification MIL-T-4734. If any handle fails to comply with the applicable requirements, it shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8834981. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Vibration (see 3.4).

4.4.1.1 Shaker-Fixture-Component Resonance Test. Prior to performance of the component vibration test, subject each component to vibration in each of its three major mutually perpendicular axes throughout the frequency range from 5-55 cps unless this has already been done with an identical component and an identical fixture on the same machine to be used for the component vibration test. Place the transit case on a suitable fixture and monitor the vibration response continuously during this test.

Note: Resonant frequencies, bandwidth and amplification of all resonances. Amplitude readings perpendicular to the direction of excitation shall be no greater than the maximum level in the direction of excitation. Alter the component-fixture-shaker combination until these conditions are satisfied. Control frequency sweep manually to insure that all resonances observed are fully excited. Vary amplitude with the frequency such that the maximum amplitude at the attachment points does not exceed the following:

Frequency	Amplitude
5-55	.03 - Peak to Peak

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If any alterations are made to the machine fixture or component, or, if a different machine or fixture is used after the above test has been made, repeat this test, incorporating the new conditions, prior to component testing.

4.4.1.2 Component test. With the dummy test set (see 6.4) enclosed in its case in its intended carrying condition (see dwg. 8834979) and mounted to the vibration machine in the manner determined by the fixture resonance test (4.4.1.1), vibrate the test set in each of 3 mutually perpendicular axes at the frequency, time and amplitude ranges specified below. After completing the vibration runs, perform the pressure test (see 4.4.5).

FREQUENCY (CPS)	SWEEP TIME (MINUTES)	DOUBLE AMPLITUDE (INCHES)	NUMBER OF SWEEPS
5-55-5	1 ± .2	.06 ± .006	15

4.4.2 Drop test (see 3.5). Subject the dummy test set (see 6.4), enclosed in its transit case in the intended carrying condition (see dwg. 8834979), to the free fall drop test specified in MIL-P-116. After completion of test perform the pressure test (see 4.4.5).

4.4.3 Low temperature (see 3.6). The transit case with its dummy test set (see 6.4) in accordance with dwg. 8834979 shall be placed in a chamber whose temperature is minus 65 ± 5 degrees fahrenheit for a minimum of 4 hours. Remove the transit case from the chamber and allow it to return to plus 80 ± 20 degrees fahrenheit without the use of forced heating. The transit case shall then be tested in accordance with 4.4.5.

4.4.4 Paint adhesion (see 3.7). Paint adhesion shall be tested in accordance with Paint Adhesion, Section 4 of TT-C-490.

4.4.5 Pressure (see 4.3.3.1). The transit case shall be tested as specified on dwg. 8834981.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The transit case shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The transit case shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The transit case shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

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b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Dummy test set. A dummy test set of the same contour and weight as the test set shall be furnished the contractor by the contracting officer.

6.5 Supersession data. This specification includes the requirements of Purchase Description PA-PD-2305, Box, Pkg, Ammo, for Test Set: T4090 and Case Transit without test set T4090 dated 14 December 1965.

Custodian:
Army - MU

Preparing activity:
Army MU

Project No. 8140-A168

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining *information on the use* of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-50228(MU) Case, Transit, for Test Set, T4090

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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